



University of Pittsburgh

Vestibular Lab Workbook

Pitt SHRS

Department of Physical Therapy
School of Health and Rehabilitation Sciences

FORWARD

The University of Pittsburgh Advanced Vestibular Physical Therapist (AVPT) team hopes that you and ultimately your clients will benefit from the vestibular physical therapy manual. The goal of the AVPT program is to take great vestibular physical therapists and make them even better. The program is designed to elevate practice and help you to become the vestibular physical therapist expert within your community. The program culminates with face-to-face practice, friendship building among the participants, and the amalgamation of a year of study via learning about complex cases and video presentations by some of the best vestibular physical therapists in the world. Our goal is to improve care globally to improve the lives of persons living with dizziness and balance impairments. We hope that all of you will become leaders in moving the science and practice of vestibular physical therapy to new heights.

Our sincere thanks go to our models including Mrs. Janice Griffiths, Mr. Lloyd Griffiths, and Ngozi Elendu, DPT. Thanks also to Wei-Neng Chuang, PT, MS our photographer, Chelsea Manetta, DPT, NCS, Amy Cassidy, DPT, NCS, and Regan Harrell, DPT, NCS of our AVPT team. We would like to acknowledge Dr. Harrell for driving the creation of this manual and Tara Ridge, PT, MS, SCS, CSCS, Vice Chair of Post-professional education in the Department of Physical Therapy for providing the funding support.

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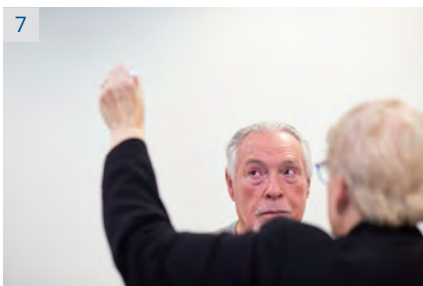
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Clinical Exam Tests

Nine Quadrant Test

The patient looks at a 9-point grid (high/middle/low X left/center/right) as the physical therapist (PT) moves a target (index finger). May need to lift eyelid for down gaze positions. The PT observes for nystagmus in any of the directions.



Gaze Holding Nystagmus

The PT asks the patient to look in eccentric gaze for up to 30 seconds while the PT observes for presence of nystagmus. The patient should look to the right about 30 degrees and hold that position, then switch to about 30 degrees to the left while looking at a target (such as the PT's fingertip or a pen). The patient should also look up and down 30 degrees. This can be combined with eye movement range of motion testing.



Smooth Pursuit

The PT holds a target (fingertip) at a distance of 3 feet from the patient. The PT asks the patient to follow the target as the PT slowly moves the target left to right, then up and down. The PT does not exceed 30 degrees from the nose for each eye. The target should be moved at a rate requiring approximately 2 seconds to go from right to left, another 2 seconds to go from left to right, and the same timing for up and down.

HORIZONTAL



VERTICAL



Horizontal and Vertical Saccades

The PT holds two single points (fingertips) horizontal at a distance of 3 feet from the patient; 1.5 feet to the right and 1.5 feet to the left of midline so that the patient must gaze 30 degrees to the left and 30 degrees to the right. The PT instructs the patient to move their eyes as quickly as possible from one point to the next. The same process is repeated up and down.

HORIZONTAL



VERTICAL



Convergence

The PT holds up a stick that is marked with a 14 point font target in front of the patient. The PT instructs the patient to watch the target with both eyes as it comes toward the patient's face and state when the target doubles. The PT records the distance from the patient's nose to where the patient says the target doubled. Scores of ≤ 5 cm are considered normal.



Cover and Uncover Test

COVER (indicates tropia if positive): The patient is seated looking straight ahead at a target approximately 10 feet away. The paddle (occluder) is placed over one eye and the PT observes for any eye movement of the uncovered eye. Note if the eye moves in or out, up or down. Then **UNCOVER**. Repeat for the other eye.

If a corrective movement is seen, it means that a misalignment was present. Inward movement means that an exotropia was present and named for the eye that moved. Outward movement means that an esotropia was present. Downward movement means a hypertropia was present. Upward movement means a hypotropia was present. A tropia can be fixed or alternating.



Cross-Cover Test

CROSS COVER (indicates phoria, if COVER test is negative): The patient is seated looking straight ahead at a target approximately 10 feet away. The occluder is held over one eye for 3–4 seconds and it is then quickly placed over the other eye. Repeat for 3–4 cycles. The PT observes the eye that that is being uncovered and notes if it moves in or out, up or down. The same movement will be seen in both eyes, so phorias are not named by side.

If a corrective movement is seen, it means that a misalignment was present. Inward movement means that an exophoria was present. Outward movement means that an esophoria was present. Downward movement means a hyperphoria was present. Upward movement means a hypophoria was present.



Head Impulse Test (Horizontal Canal)

The patient is instructed to look at a target in midline. The PT slowly moves the patient's head in the yaw plane about 15–20 degrees and then quickly moves the patient's head to midline (the impulse) while the patient attempts to keep their eyes on the target. The head is rapidly stopped, and the PT checks to see if the patient has a “catch-up” saccade that will refixate their eyes on the target. The head can be moved away from midline for the impulse, but it is better to move toward the midline. The movement must be fully stopped before doing the next impulse.

A normal VOR response to rapid passive head rotation is that the patient is able to fixate on the central target without saccades. An abnormal response occurs when head is rotated to the side of the vestibular lesion and there is an inability to maintain fixation during rotation of the head requiring a corrective saccade once the head has stopped moving.



HINTS Exam

The HINTS exam can be used as a tool to identify a posterior circulation stroke. The cluster of tests includes **H**orizontal head **I**mpulse Testing; **D**irection changing **N**ystagmus; **V**ertical **T**est of **S**kew. A single central finding on any of the three components rules in a posterior circulation stroke.



Head Shake Test (Head Shaking Nystagmus)

The PT tilts the patient's head forward 20–30 degrees and shakes the head in the yaw plane with eyes closed for 20 cycles quickly. The PT has the patient open their eyes and observes for post shake nystagmus. Goggles are preferred for this test.

WITH GOGGLES—PREFERRED



WITHOUT GOGGLES



Dynamic Visual Acuity Test

The patient sits at a distance that is marked at the top of the LogMAR Eye chart. The PT instructs the patient to read the LogMAR Eye chart first with the head still, and notes the lowest complete line (Static Visual Acuity). The PT then shakes the patient's head 20–30 degrees in the yaw plane with head flexed 30 degrees at 2 Hz (240 beats on a metronome, head is in right or left position with each beat). The PT notes the lowest complete line (Dynamic Visual Acuity). A positive test occurs if the patient lost more than two lines while moving the head versus while the head is still and indicates a deficit in the functional VOR.



Vibration Induced Nystagmus

While the patient is sitting with goggles donned, vibration with a frequency of 100Hz is applied to the mastoid process. In someone with unilateral vestibular loss, the resulting nystagmus will beat away from the affected ear.



Rinne and Weber Tests



Rinne test: Used to discern sound transmitted through air versus bone and helps assess for unilateral hearing loss. Using a 512-Hz tuning fork, the PT strikes the fork inducing vibration. The stem of the tuning fork is placed on the mastoid process of the patient. The patient is then asked to cover their other ear with their hand. The patient informs the PT when the sound can no longer be heard. The process is then repeated with the tines of the tuning fork held near the external auditory canal and the patient again informs the examiner when the sound can no longer be heard.

Results: Normal hearing is observed when the vibrating tuning fork positioned near the ear is louder and lasts twice as long compared to when placed on the mastoid.



Abnormal: When the sound is at least equally loud or louder when the fork is placed on the mastoid compared to when it is held next to the ear.

Weber test: Used to assess for a unilateral hearing loss and to characterize conductive versus sensorineural loss. Using a 512-Hz tuning fork, the PT strikes the fork tines to induce vibration. The stem of the tuning fork is then placed on the top of the patient's head. The patient is asked "is the sound louder in your right ear, left ear, or the same in both?"

Results: Normal hearing is confirmed when the sound is heard midline and equally on both sides

Unilateral sensorineural hearing loss is observed if the patient hears the sound louder (lateralizes) in the unaffected ear.

Unilateral conductive hearing loss is observed if the patient hears the sound louder (lateralizes) in the affected ear.

Symmetrical conductive hearing loss is observed if the patient does not exhibit lateralization.



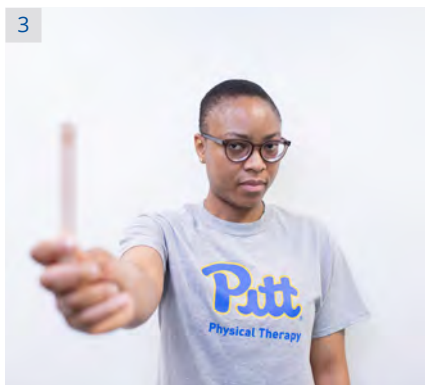
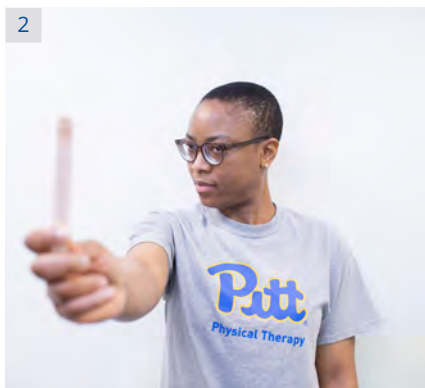
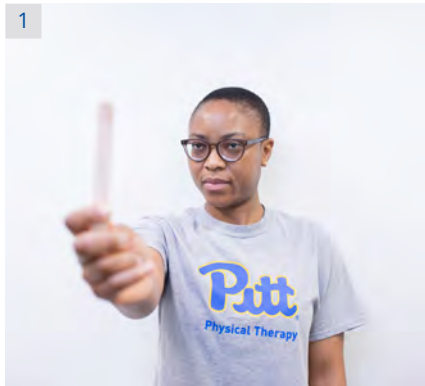


Clinical Treatment

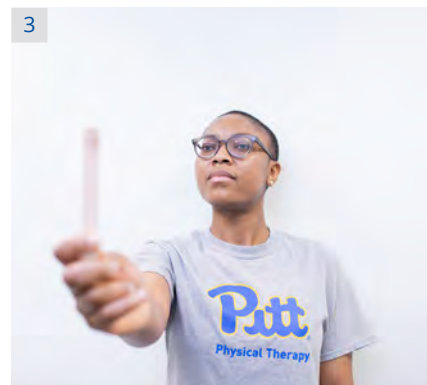
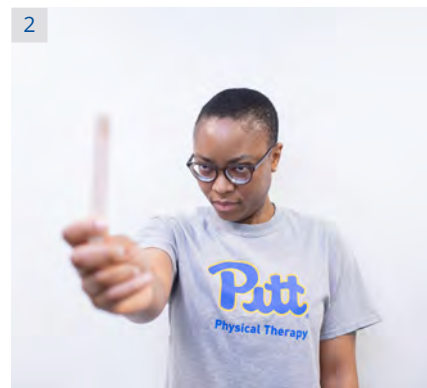
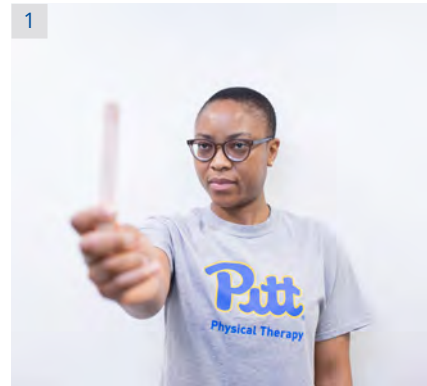
VOR x1 (x1 viewing)

The patient looks at a target about 18–24 inches away from their eyes. They are asked to move their head in either pitch or yaw planes while keeping the target in focus. A 14 point font is most commonly used for the 18 inch distance, and larger for farther away. The patient is instructed to move their head as quickly as possible but to keep the target in focus. The PT determines if the head movements increase their symptoms and at what speed. They should be able to move their head at about 100–120 beats per minute on a metronome without symptoms.

HORIZONTAL



VERTICAL

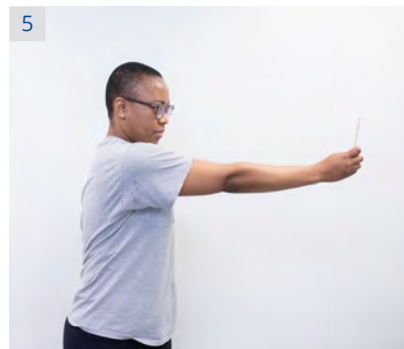
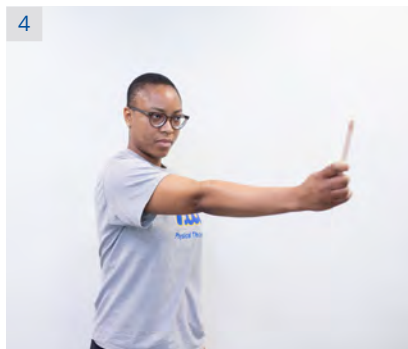
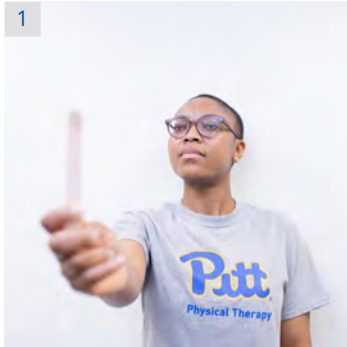


VOR Cancellation (Visual Motion Sensitivity)

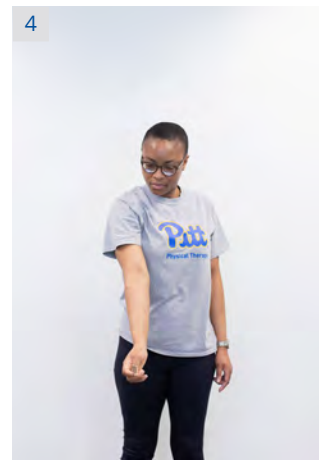
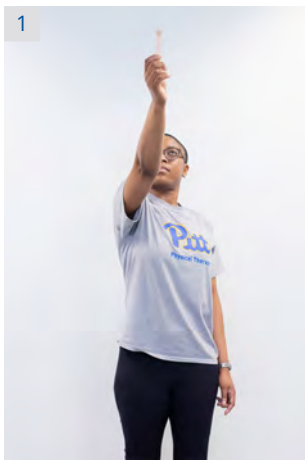
The patient extends their arm directly in front of their body or clasps their hands together. The patient is instructed to visually fixate on the target and maintain fixation. The entire body is then rotated 80 degrees back and forth to each side at a comfortable speed for several cycles (use a metronome at 50 beats/minute). May use a swivel chair/stool or the patient can twist from waist using a stationary seat or stand with feet apart. NOTE: Patient must keep the head, neck and shoulders moving together as a unit and their hands/thumbs straight out in front of their body.

A normal response is when the eyes remain stationary and motionless throughout the test.

HORIZONTAL

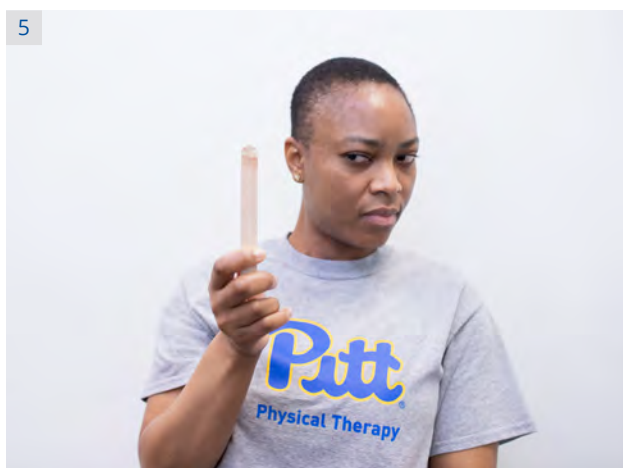


VERTICAL



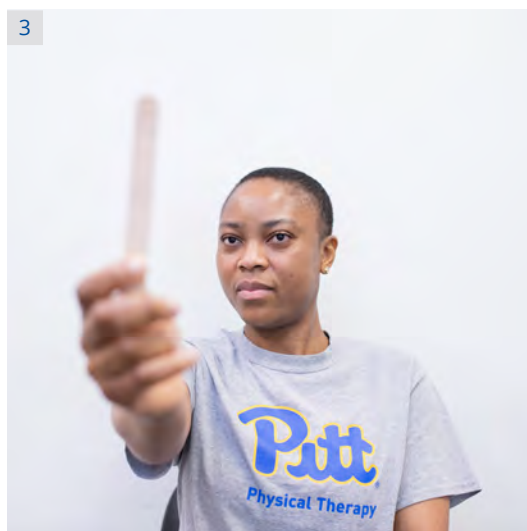
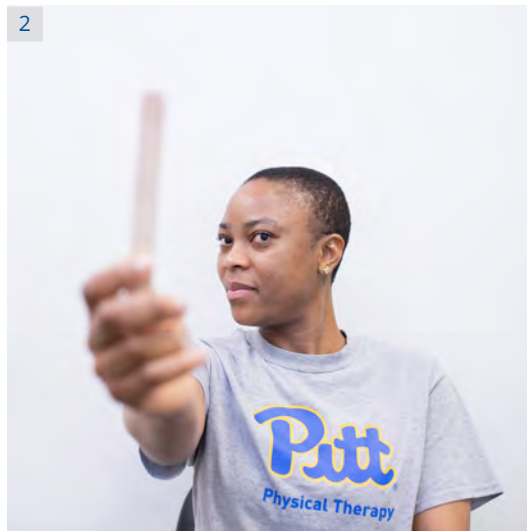
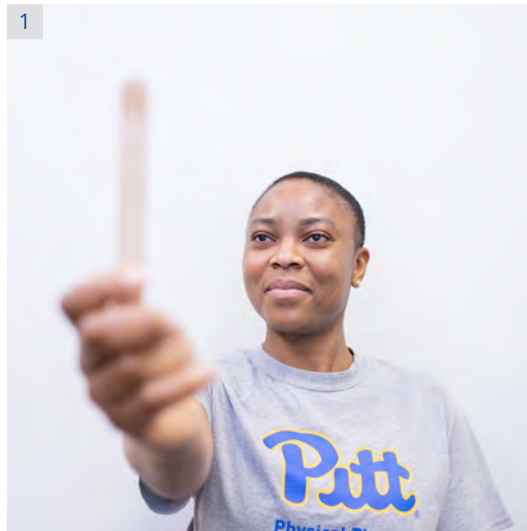
VOR Plus Vergence

The PT instructs the patient to perform the VORx1 exercise while moving the target toward and away from their nose so that they are doing vergence and the VORx1 exercise simultaneously.



Head Impulse Exercise

The patient is seated and focusing on a target about 3 feet from their nose. They quickly move their head about 15–20 degrees toward the affected side and then slowly move their head back to center.



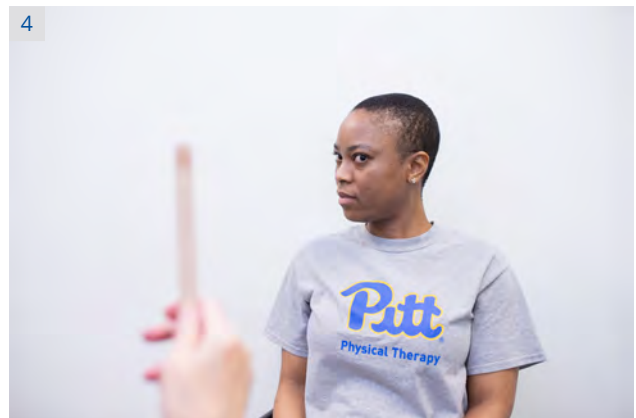
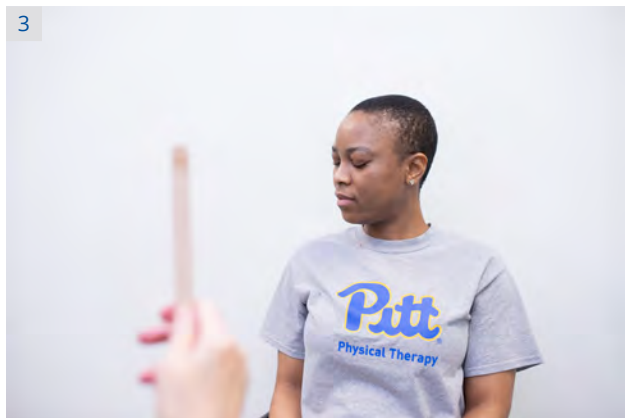
Eye Head Saccade Exercises

This exercise is prescribed for patients with unilateral and bilateral loss but definitely for patients with bilateral loss. There are two targets in front of them at least 18 inches away at about a 30 degree angle from their nose to the right and the left (or up and down). The patient is to look at one target, and then move their eyes ONLY to the second target, then move their head and repeat this in the opposite direction. The goal is to attempt to have the saccadic eye system help to compensate for the damage to the VOR.



Imaginary Targets (Remembered Targets)

The PT has the patient look at a target directly in front of them. The PT instructs the patient to close their eyes and turn their head slightly in the yaw plane imagining that they are still looking at the target. The PT instructs the patient to open their eyes and see if they were able to keep their eyes on the target. The patient repeats the exercise in the opposite direction. The patient should NOT be holding the target.



Brock String



The patient holds one end of the Brock string at the tip of their nose and the other end is attached to a stationary object so that the string is taut. Move the nearest bead to the near point of convergence or just closer. Move the far bead to arm's length and place the middle bead at the midpoint of the other two. The patient starts by focusing on the far bead and holds for 5–10 seconds. The patient should report seeing two lines diverging toward them like the letter A from the far bead.

Next, the patient changes focus to the middle bead and holds for 5–10 seconds. The patient should report seeing two lines diverging toward and away from them like the letter X from the middle bead.

Then the patient changes focus to the close bead and holds for 5–10 seconds. The patient should report seeing two lines diverging away from them like the letter V from the close bead.

Then the patient moves focus back to the middle bead and far bead, holding each for 5–10 seconds. This should be repeated until symptoms increase by several units or the patient needs to take a break. Progress by changing starting point, increasing reps, or adding more beads to the string.



Posterior Canal BPPV Positional Tests

Dix-Hallpike

After checking cervical range of motion, with the patient in long-sitting position on a mat table, the PT places patient's head in 45 degrees of yaw rotation. The PT quickly (1–2 seconds) lowers the patient to supine position with head extended 30 degrees below the edge of the mat. The PT observes for nystagmus to become evident over 30 seconds and notes the direction of the nystagmus. For posterior canal BPPV, nystagmus should be upbeating and torsional toward the tested side. Good body mechanics are utilized to avoid injury. The PT maintains contact with the patient after bringing the patient back up to the sitting posture and is careful throughout the maneuver to avoid neck or back pain for the patient and back pain for the PT.

WITHOUT GOGGLES



WITH GOGGLES—PREFERRED



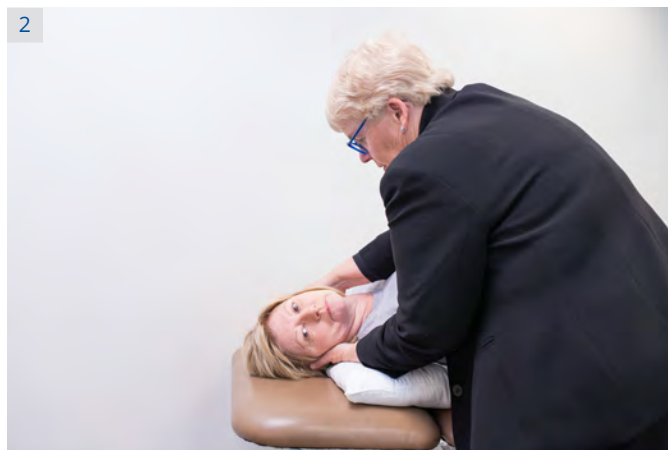
Loaded Dix-Hallpike

Perform the same as Dix-Hallpike, except begin with head rotated 45 degrees towards the side being tested, flexed 30 degrees to chest, and while in the long sitting position (i.e., pre-loading the head). Hold for 30 seconds before placing the patient in supine.



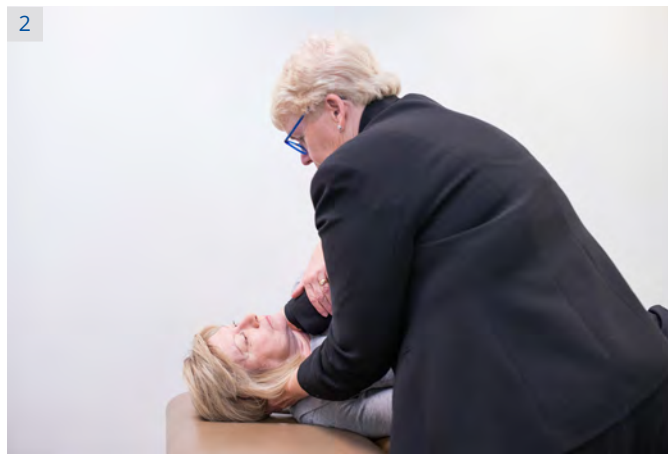
Modified Dix-Hallpike Test With Head Over Pillow

The patient, with their head rotated in yaw 45 degrees, is laid down in supine with their head extended over the edge of a pillow approximately 20–30 degrees in extension. Care must be taken to choose the correct pillow size in order to avoid excessive extension.



Sidelying Test

The patient starts in sitting with legs resting over edge of table (hips/knees at 90 degrees). The PT has the patient turn their head 45 degrees away from the ear being tested and brings them rapidly down sidelying to the table on the side being tested, with the nose toward the ceiling. The PT observes for nystagmus to become evident over 30 seconds and notes the direction of the nystagmus. The PT understands what a positive test indicates when visualized. The PT brings the patient back up to sitting, taking care to hold onto the patient as they return to sitting and for about 30 seconds to 1 minute after they have returned to sitting.



An orange textured vertical bar is located on the left side of the slide, extending from the top to the bottom.

Posterior Canal Canalith Repositioning Maneuvers

Modified Epley

The PT instructs and demonstrates for the patient and explains the purpose of the maneuver.

STEP 1: The PT seats the patient on the table in long sitting with the head in 45 degrees rotation toward the affected ear. The PT quickly lowers the patient to supine (1–2 seconds) so they may be taken back to the head-hanging position with the neck in slight extension (20–30 degrees of extension). The PT stabilizes the head with their hands and waits for the nystagmus to stop. The PT observes for eye movement in the head hanging position.

STEP 2: The PT rotates the head 90 degrees towards the uninvolved side by turning the head toward opposite side. The PT observes the patient's eyes and waits for nystagmus to stop. Keep the neck in extension during the turn to the opposite side and wait 30 seconds.

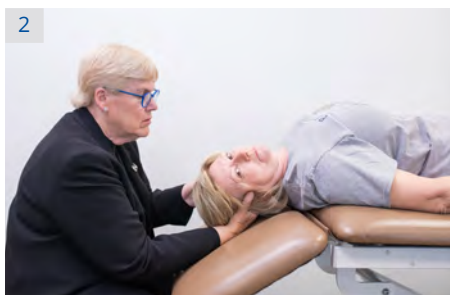
STEP 3: Log roll patient onto unaffected side so that the head is turned to face ground. Hold and wait 30 seconds.

STEP 4: The patient keeps their chin tucked towards their shoulder and rises to sitting while the PT HOLDS ON TO THE PATIENT with both hands. The PT slowly turns the patient's head back to midline. The PT HOLDS ON TO THE PATIENT as they come to sitting and up to a minute or more after completion of the maneuver.

WITH GOGGLES—PREFERRED



WITHOUT GOGGLES



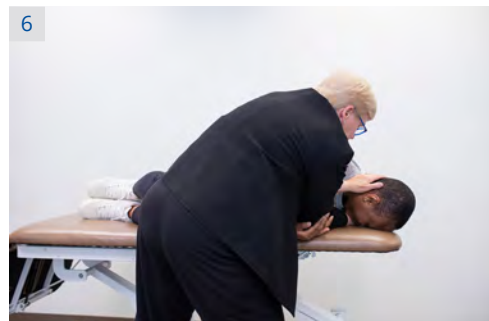
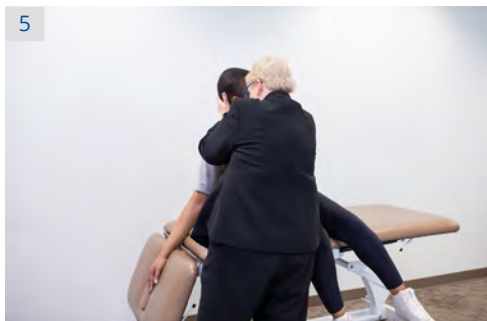
Semont

The PT rotates the patient's head 45 degrees away from the affected side and quickly moves the patient into side-lying (head now facing up, image 3). Stay in this position for 1–2 minutes. Quickly, in one motion, sit up and move the patient onto the opposite side-lying (good side) without changing head position (nose now facing the ground, image 6). Stay in this position for 1–2 minutes. Remember that modeling suggests the 'flip' needs to be completed in <1.5 seconds. The PT then returns the patient slowly to sitting. If reversal of nystagmus occurs (or unsure), repeat up to three repetitions.



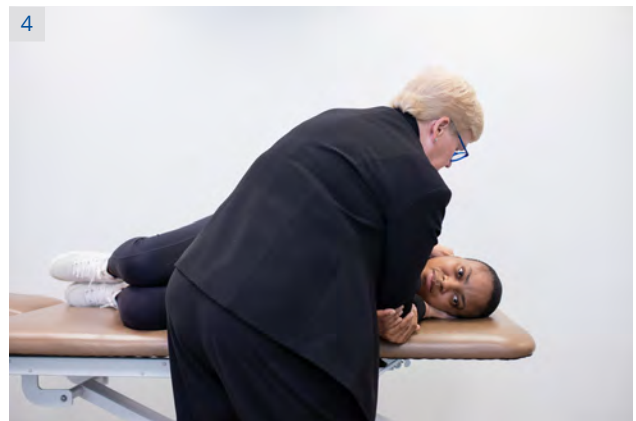
Semont Plus

The patient begins sitting upright and facing forward, then the PT turns the patient's head away from the affected ear. The PT quickly lays the patient down on the side of the involved ear, over the edge of the table about 20–30 degrees so that the patient's dependent arm can abduct off the table and rest on the stool. The patient stays there until the vertigo passes. The PT QUICKLY sits the patient up and then lays them on their other side while KEEPING the patient's head pointing over the shoulder the entire time (the orientation of the head should not change). This should be completed in one movement without stopping in the middle. The patient should be facing down toward the ground. Wait there until any vertigo/nystagmus passes. The PT then HOLDS ON TO THE PATIENT as they come to sitting, and continues holding onto them for up to a minute or more after completion of the maneuver.



Li Maneuver

Start with patient in sitting with legs hanging over edge of the table (hips/knees at 90 degrees). The PT moves the patient down quickly to their involved side with their nose forward while holding their head (no turning of the head is required for this maneuver). After 1 minute, or after resolution of the nystagmus, the PT rapidly moves the patient 180 degrees with the healthy ear down on the table and their nose facing forwards. The patient lies there for 4 minutes and then the PT rapidly lifts them back to sitting. The PT HOLDS ON TO THE PATIENT for up to several minutes after completion of the Li maneuver.





Horizontal Canal BPPV Positional Tests

Long Sit to Supine

The patient starts in long sitting and then moves to supine quickly, ending in approximately 30 degrees of neck flexion (on a pillow will usually work fine). The PT observes eye movement if there is any.

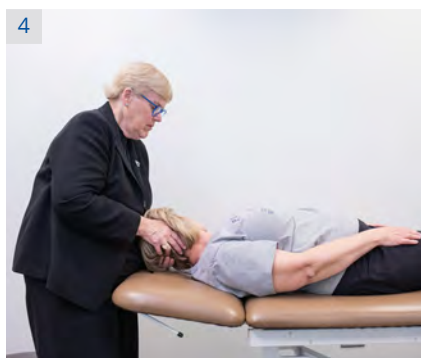
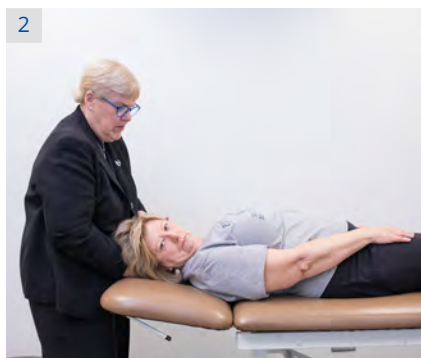


Roll Test (Pagnini-McClure Roll Test)

WITH GOGGLES—PREFERRED



WITHOUT GOGGLES



The patient starts supine with their head flexed 20-30 degrees. The PT quickly moves the head to the right and holds for 30 seconds. The PT observes for nystagmus to become evident over 30 seconds and notes the direction of the nystagmus. The PT moves the head back to midline and waits for 30 seconds before quickly moving the head to the left and holding for 30 seconds. The PT observes for nystagmus to become evident over 30 seconds and notes the direction of the nystagmus. If positive, nystagmus should be beating towards the ground (geotropic), OR away from the ground (ageotropic), in both right and left head positions. The PT should note which side the patient experienced the worse symptoms and which side had the more intense nystagmus.

Bow and Lean Test

The patient begins in sitting. BOW: The patient flexes their head to be parallel to floor and holds for 30 seconds. The PT notes the direction of nystagmus. If geotropic nystagmus is observed with the roll test, the nystagmus will beat towards the affected ear. If ageotropic nystagmus is observed with the roll test, the nystagmus will beat towards the healthy ear.

LEAN: The patient extends their head as if looking at ceiling for 30 seconds. The PT notes the direction of nystagmus, which should be opposite to what you saw in the BOW test. If geotropic nystagmus is seen with the roll test, the nystagmus will beat towards the healthy ear. If ageotropic nystagmus is seen with the roll test, the nystagmus will beat towards the affected ear.
(Geotropic → nystagmus beats toward Good, Ageotropic → Affected)

Results from this test will be used to confirm the side of the affected ear if there is uncertainty about which ear had the greatest nystagmus/vertigo.

WITH GOGGLES—PREFERRED



WITHOUT GOGGLES

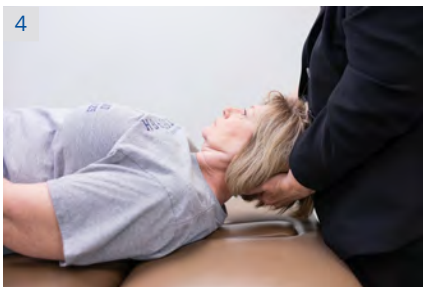


A vertical orange bar with a textured, marbled appearance is located on the left side of the slide.

Horizontal Canal Canalith Repositioning Maneuvers

Barbecue Maneuver for Canalithiasis

The patient begins in a supine position with their head flexed 30 degrees. The PT then rotates the patient's head 90 degrees towards the affected side; this position is held for 30 seconds after the nystagmus stops. From here the PT rotates the patient's head back to neutral. Next the patient's head is rotated 90 degrees towards the unaffected side. Again this position is held for 30 seconds after the nystagmus stops. The patient then rolls prone, after 30 seconds from when the nystagmus stops the patient can either get up from the prone position or complete the roll and return to sitting from the sidelying position.



Gufoni Maneuver for Canalithiasis

The patient sits in the middle of a mat table with legs hanging over the side and the head in midline. The PT briskly moves the patient down onto the UNAFFECTED side and holds for 30 seconds. The head is then quickly turned down by 45 degrees and held there for 2–3 minutes, then the patient is brought back into the sitting position.



Li Quick Maneuver for Canalithiasis

The patient starts in supine in the middle of a large bed. The PT log rolls the patient 90 degrees onto the affected side and waits until the symptoms resolve. The PT then stands behind the patient and grabs the patient's hand on the involved side and QUICKLY log rolls them 180 degrees onto their unaffected side. The patient stays on the unaffected left side for 2 minutes and then the PT brings the patient to sitting. The maneuver takes approximately 3 minutes to perform.



Gufoni Maneuver for Cupulolithiasis

The patient sits in the middle of the table with legs hanging over the side and head in midline. The PT instructs the patient to look straight ahead, then brings the patient down onto their AFFECTED side for 30 seconds. Then the PT turns the patient's head up towards the ceiling and lets them rest in this position approximately 2 minutes. At the end of the 2 minutes, the PT brings the patient back to sitting, holding onto the patient throughout. If this does not work, it is common to try the same maneuver but with the patient's nose tilted down towards the floor.



Kim Maneuver for Cupulolithiasis

Starting with the patient in supine, the PT log rolls the patient onto the affected side, then turns the patient's head 135 degrees towards the floor. The PT places the vibrator over the mastoid of the affected side for 30 seconds and waits for the nystagmus to stop. The PT log rolls the patient back to side lying (90 degrees from supine), then log rolls the patient to supine nose up position. Next the PT logs rolls the patient 90 degrees to the unaffected side and vibrates over the mastoid again for 30 seconds. Finally log roll the patient to prone. The PT will move the patient carefully and ensure that they do not excessively pull on the cervical spine.

